

Darren Biskup

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EDUCATION

Columbia University

M.S./Ph.D. Mechanical Engineering (Robotics & Rehabilitation)

Columbia-Dream Sports AI Ph.D. Fellowship

New York, NY

Aug 2024 - Present

GPA: 4.00/4.33

University of Illinois, Urbana-Champaign

B.S. Mechanical Engineering, Computer Science Minor

High Honors, James Scholar Honors Program, Dean's List

Urbana, IL

May 2024

GPA: 3.95

SKILLS

Languages & Software: C/C++, Python (PyTorch, OpenCV), CATIA 3DEXPERIENCE, SolidWorks, ROS2, Git

Controls & Estimation: Model Predictive Control, Trajectory Optimization, Sensor Fusion (EKF/UKF), MuJoCo

Hardware & Design: CAN/CAN-FD BLDC Control, Injection Molding, GD&T, RTOS,

EXPERIENCE

Robotics And Rehabilitation (ROAR) Lab, Columbia University

Aug 2024 - Present

Graduate Research Assistant

- Engineered the hierarchical control architecture for a cable-driven physical assistance robot, integrating VICON motion capture and force plate sensing. Closes loop on desired net body wrench by synchronizing multi-axis motor commands via CAN bus with inline load cell tension feedback.
- Authored IEEE RA-L submission detailing a real-time, Ground Reaction Force-aware convex MPC leveraging human centroidal dynamics for underactuated upright assistance. Validated against a fully-actuated impedance baseline, the architecture reduced dynamic tracking error by 24% and mechanical work by 29%.
- Designing force guidance protocols for dynamic yoga pose training based on comparative analysis of experimentally collected expert-novice movement data, leveraging automated kinematic assessment.

Mehta Research Group/GazzolaLab, UIUC

Jan 2023 - May 2024

Undergraduate Research Assistant - Soft Robotics

- Connected fiber-reinforced soft actuators in parallel and implemented joystick control for the robotic soft arm using a Raspberry Pi to control pneumatic solenoid valves.
- Implemented a recursive low-pass-filter to enhance joystick input in real-time.
- Designed and built a 3D calibration grid using laser-cut acrylic and green paint; developed Python code utilizing OpenCV and scikit-image to detect colored blobs and extract marker coordinates.
- Developed scripts to input arm pose maneuver instructions to allow rigorous comparison between simulation and real-world.

Skydio

May 2022 - Aug 2022

Product Design Engineering Intern

- Design for Injection Molding: designed the Navigation Camera mounting mechanism for the company's next generation performance quadcopter drone. Communicated with overseas vendors in China to implement design change, ensuring injection moldability and minimal lead time.
- Designed and prototyped the mobile tablet adapter for the next generation drone controller using FDM and SLS 3D-Printing. Received DFM feedback from injection molding vendor.
- Worked with Manufacturing Engineers to design wire routing layout for drone RF cables, board to board connections, 3-phase motor power cables, LED cables.

Lucid Motors

May 2021 - Aug 2021

Mechanical Engineering Intern, High Voltage

- Battery Pack Development: Improved design for the high voltage chain halving the number of bolts required to join busbars. Conducted thermal analysis on busbar joints with new design to evaluate the new design's effect on car's horsepower, range, efficiency, and thermal endurance.
- Collected and analyzed data on heat generation from bolted busbars joints, and used this data to calculate theoretical horsepower and efficiency loss of specific joints.

PROJECTS

Senior Capstone Project - Neurosurgical Balloon Tunnelling Device

- Developed an innovative shunt passage method using expandable balloons for ventriculoperitoneal operations which minimizes tissue damage and backtracking leading to improved patient and surgeon comfort.